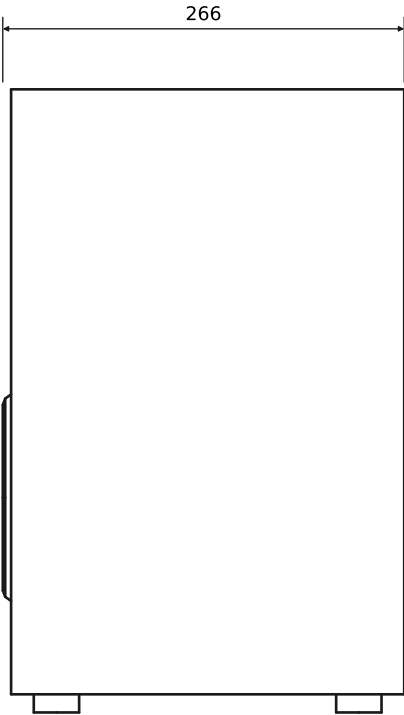
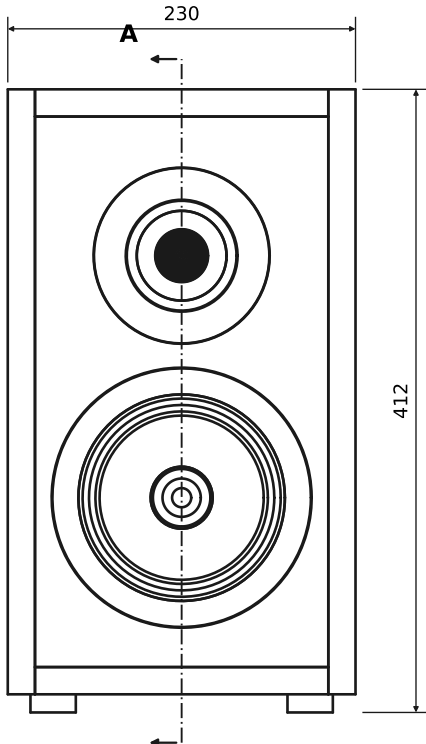


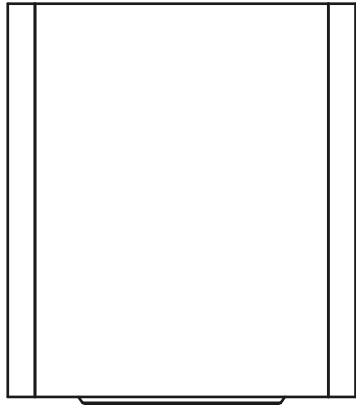
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT A



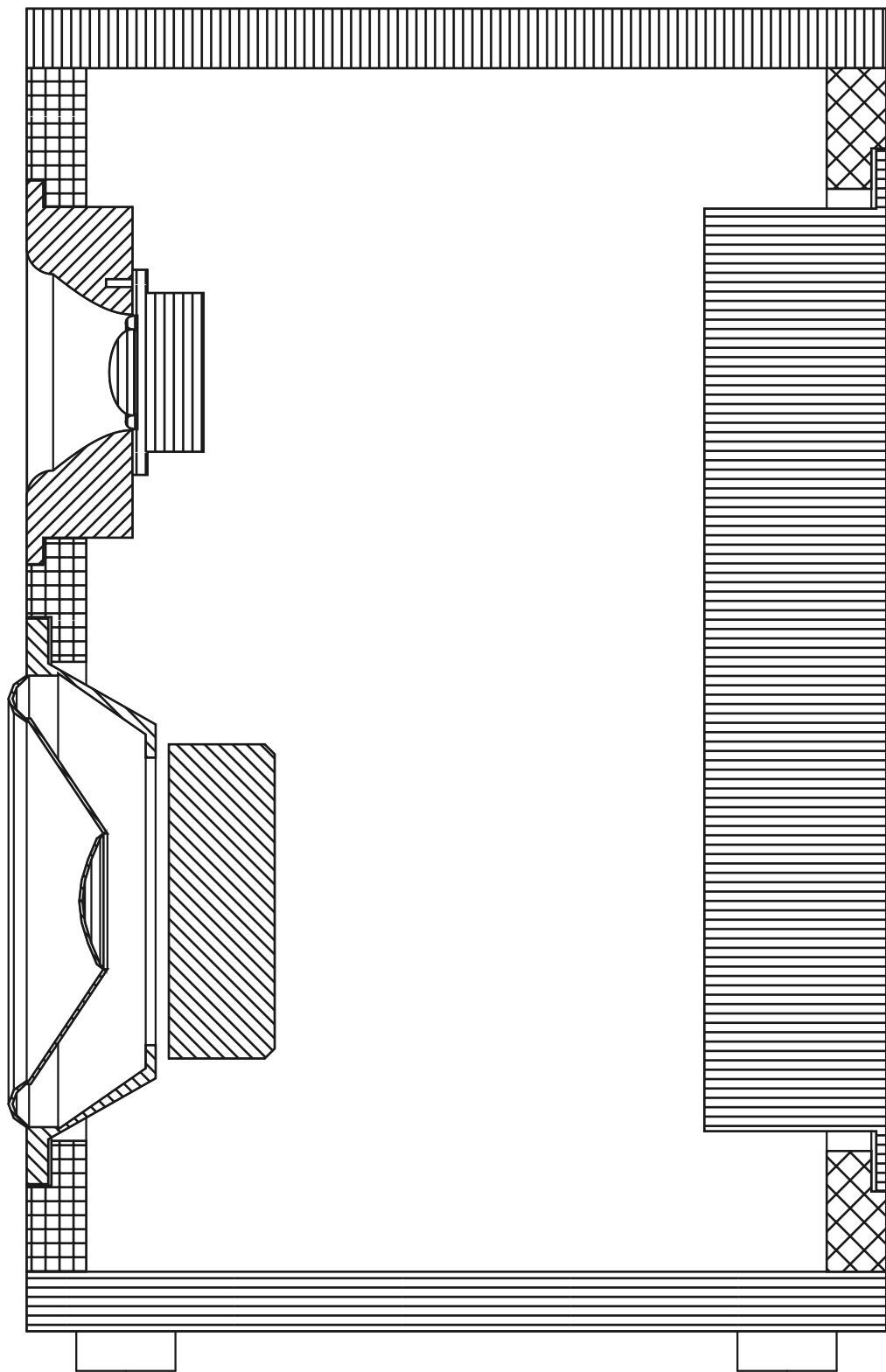
FROM ABOVE

General arrangement

Birch Box: Two-way active bookshelf speaker (the kit's example)

mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x 115.0 and seen from the right; cut faces hatched, a different angle for each part.

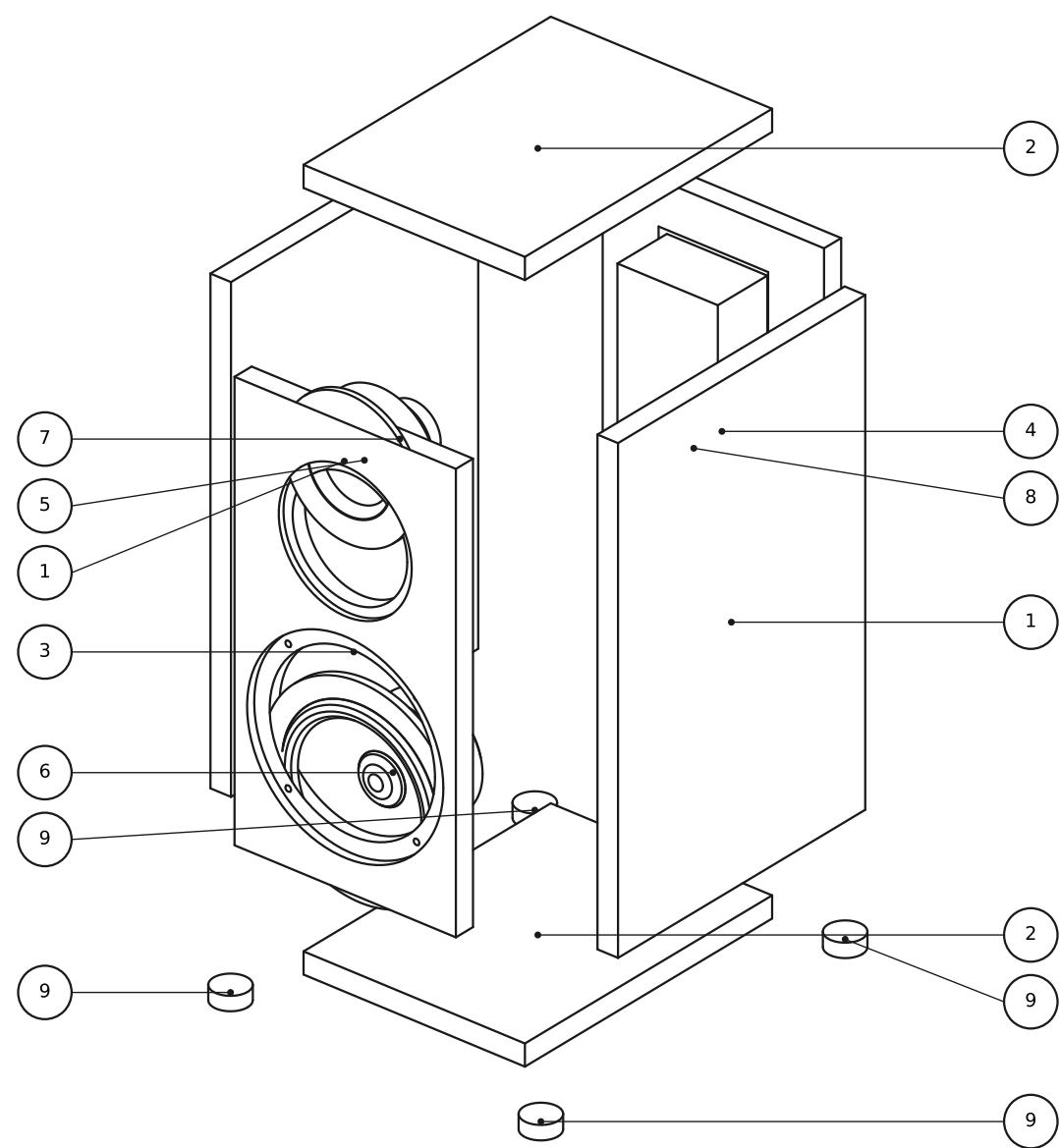


Section A-A

Birch Box: Two-way active bookshelf speaker (the kit's example)

mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).

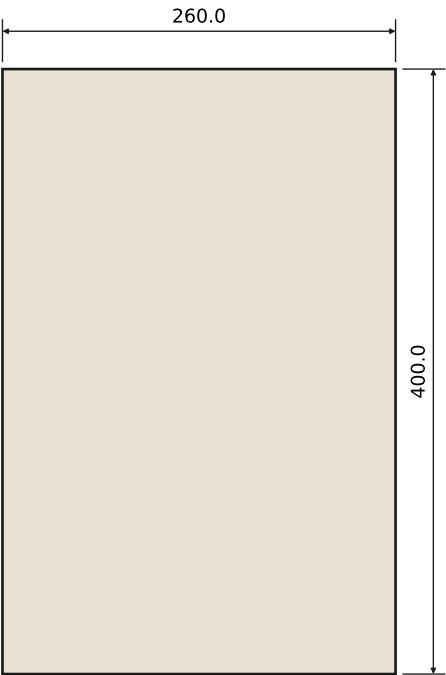


Exploded view

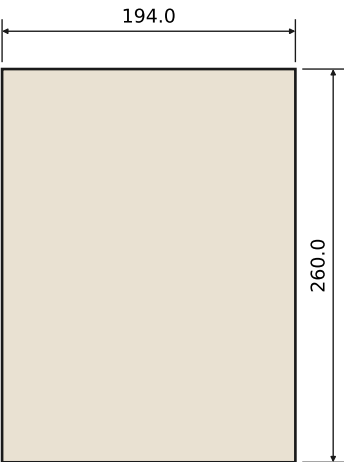
Birch Box: Two-way active bookshelf speaker (the kit's example)

mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



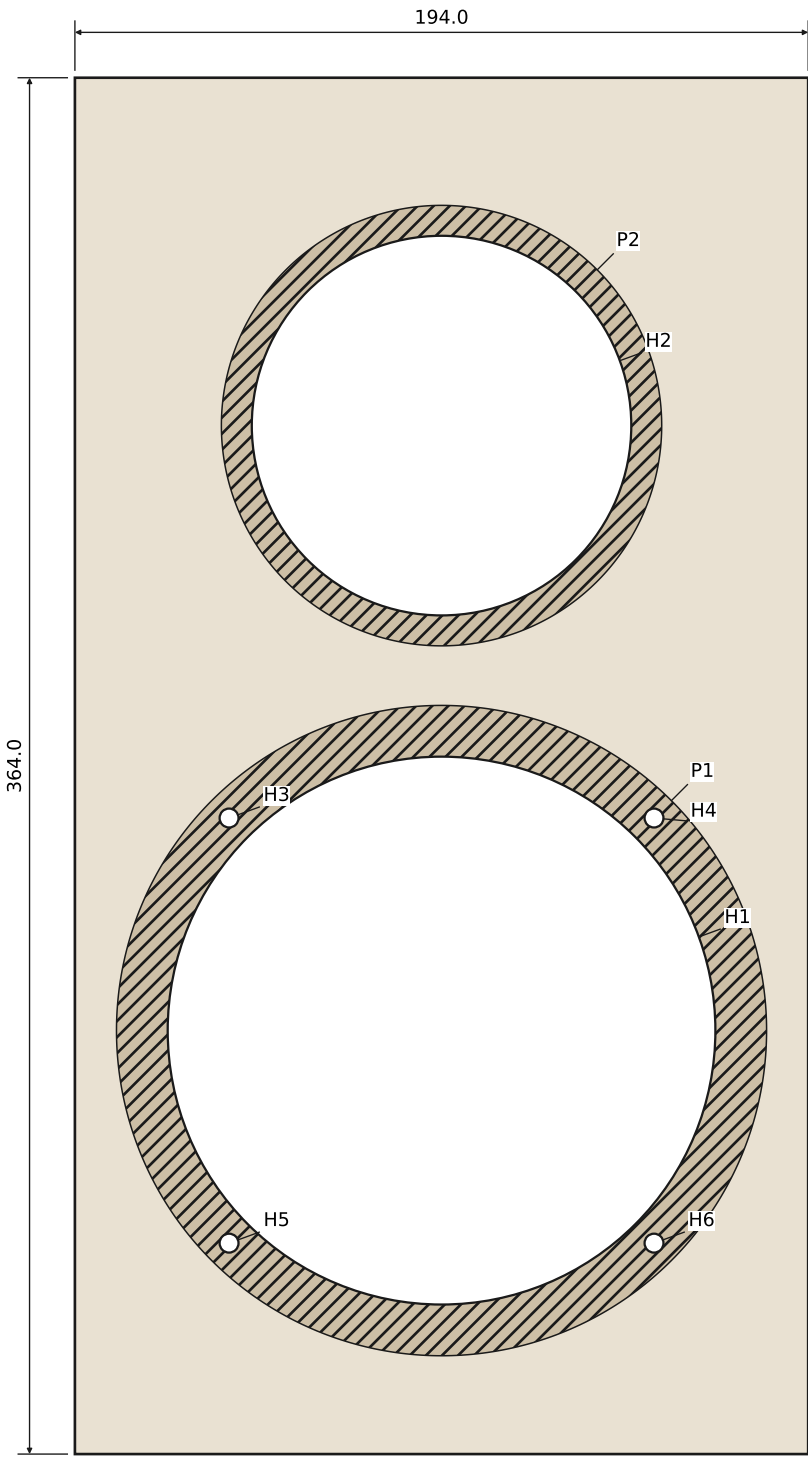
left, right
18 mm baltic-birch, 4 for the set



top, bottom
18 mm baltic-birch, 4 for the set

Sheet parts: plain panels	
Birch Box: Two-way active bookshelf speaker (the kit's example)	
mm issue 2026-10-11 first angle Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15	4 of 8 BIR-004 rev A

Scale 1:2. As the router sees it from the outer face; DXF: dxf/baffle.dxf. Marks listed at the right.



baffle: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø172.0	97.0	112.0	7.5 from face
P2	Ø116.5	97.0	272.0	5 from face
H1	Ø144.9	97.0	112.0	through
H2	Ø100.4	97.0	272.0	through
H3	Ø5.0	40.8	168.2	through
H4	Ø5.0	153.2	168.2	through
H5	Ø5.0	40.8	55.8	through
H6	Ø5.0	153.2	55.8	through

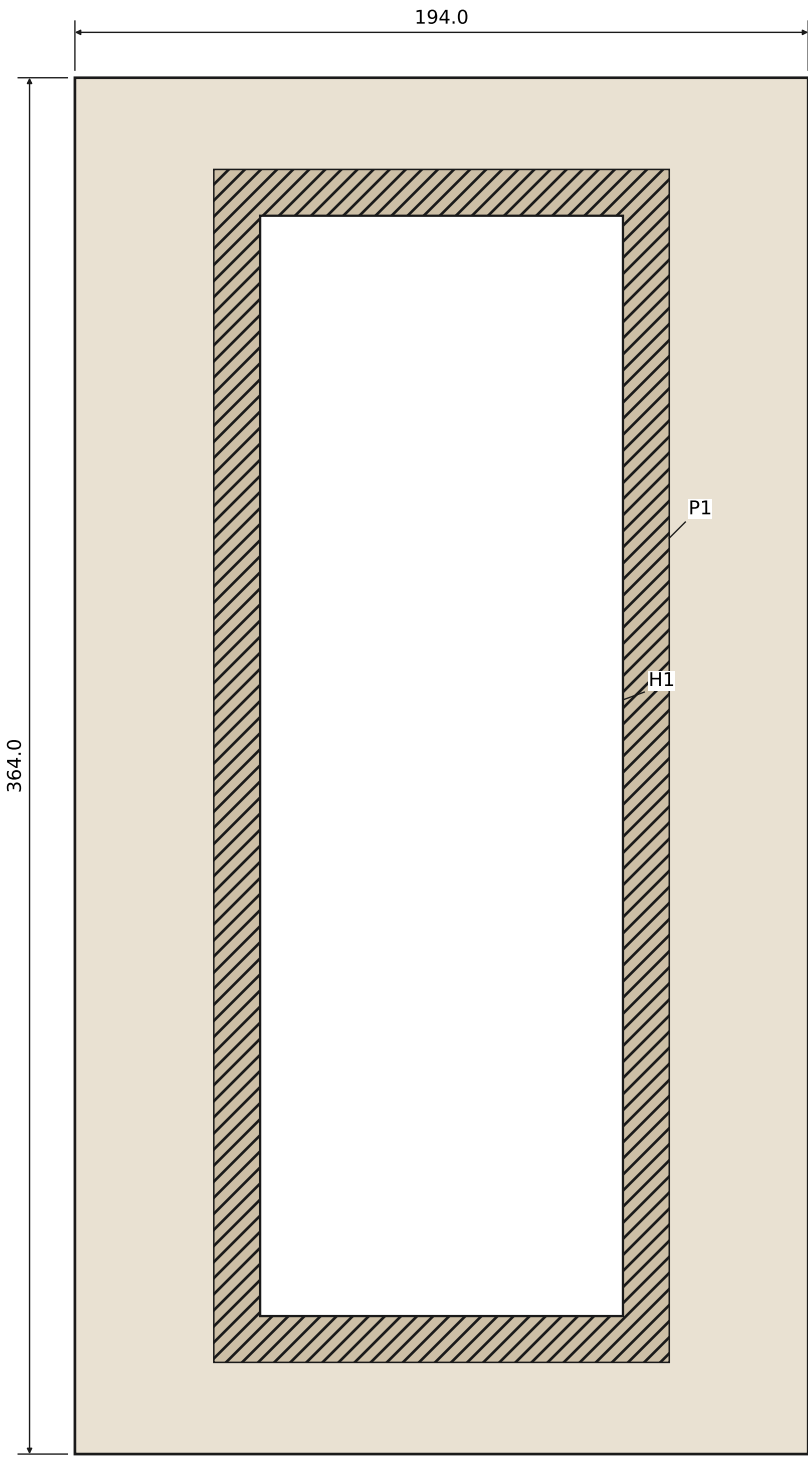
H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: baffle

Birch Box: Two-way active bookshelf speaker (the kit's example)

mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	97.0	182.0	4.5 from face
H1	96.0 x 291.0	97.0	182.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

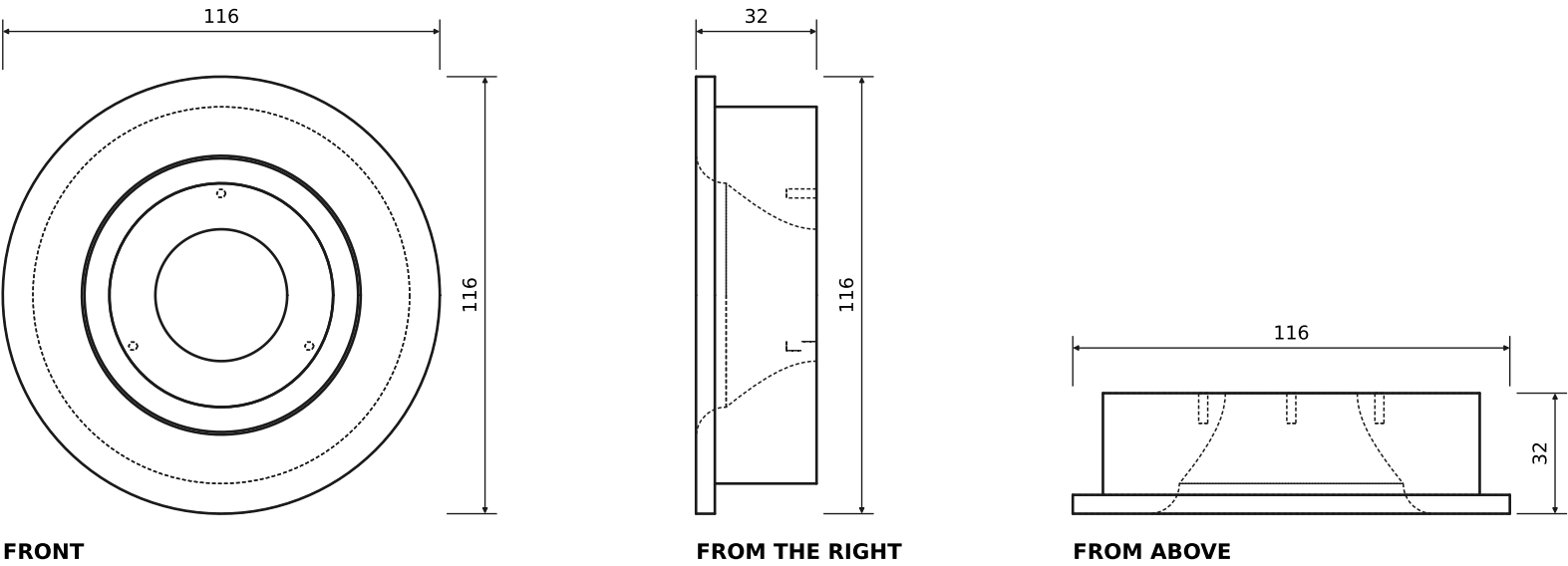
Sheet part: back

Birch Box: Two-way active bookshelf speaker (the kit's example)

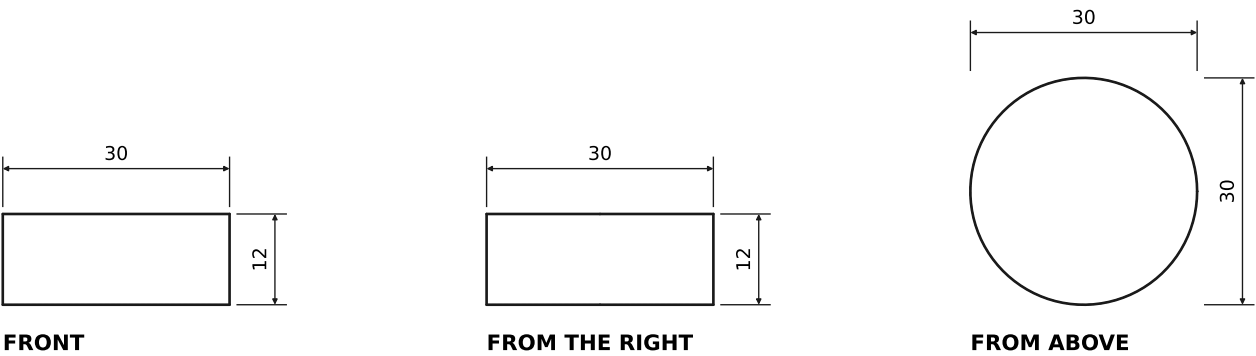
mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. waveguide: resin-tough print, front face down; 202 cm3 each; 2 for the set; 1:2
oblate-spheroidal, r0 17.5, 90 degrees, mouth 74 across



9. foot-1, foot-2, foot-3, foot-4: lathe, oak; 8 cm3 each; 8 for the set; 1:1



Printed and machined parts

Birch Box: Two-way active bookshelf speaker (the kit's example)

mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	260 x 400	7.54
2	top, bottom	cut	4	18 mm Baltic birch plywood, BB/BB	194 x 260	3.66
3	baffle	cut	2	18 mm Baltic birch plywood, BB/BB	194 x 364	5.12
4	back	cut	2	18 mm Baltic birch plywood, BB/BB	194 x 364	5.12
5	waveguide	print	2	Tough (ABS-like) SLA resin	202 cm3	78.82
6	woofer	buy	2	SB Acoustics SB17NRX2C35-8 6.5 in Norex (paper cone)	171 frame, 144.9 cut-out, 75 deep	76.70
7	tweeter	buy	2	Scan-Speak Illuminator D3004/602200 26 mm dome	62 faceplate	165.00
8	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
9	foot-1, foot-2, foot-3, foot-4	lathe	8	White oak, solid	40 x 40 x 22	10.63
10	woofer-fixings	buy	2	M4 x 20 low-profile button heads (A2) with M4 T-nuts, a driver's set of 4	M4	4.00
11	gasket	buy	2	1.5 mm closed-cell foam gasket strip, 1 m		5.00
12	damping	buy	2	Polyester fibre fill, 150 g		8.00
13	amp-fixings	buy	2	Hypex 4.3 x 16 self-tapping pan heads and 3 mm EPDM tape, a plate's set		10.00
14	cable	buy	2	Round-sheathed speaker cable 2 x 1.5 mm2, 1 m		2.00
15	terminals	buy	8	6.3 mm insulated push-on terminal		0.50

Notes

- 1. Glue-up: sides to the bottom, then the baffle and back, then the top; wood glue, clamped 1 h each.
- 2. Fit the drivers on their gaskets with the T-nuts hammered in from inside before the back goes on.
- 3. The amplifier's screw holes are drilled from the plate in hand (Hypex does not publish the pattern).
- 4. Hypex advises a sealed box round the module; this example leaves it out for clarity.
- 5. Oil the plywood; print the waveguide in black or spray it satin black.

Parts list and notes

Birch Box: Two-way active bookshelf speaker (the kit's example)

mm | issue 2026-10-11 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15